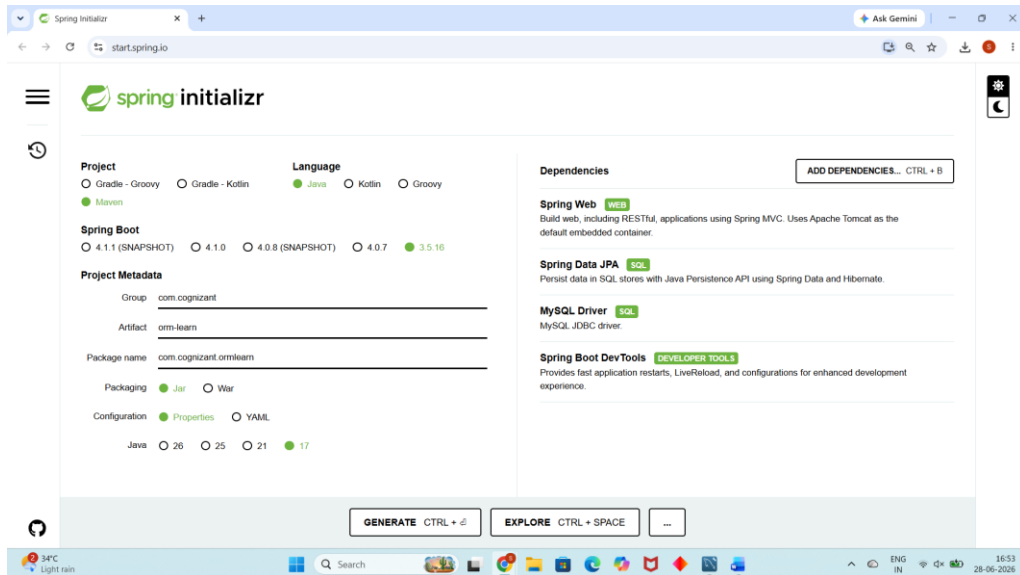


Spring Data JPA with Spring Boot, Hibernate

Spring Data JPA - Quick Example

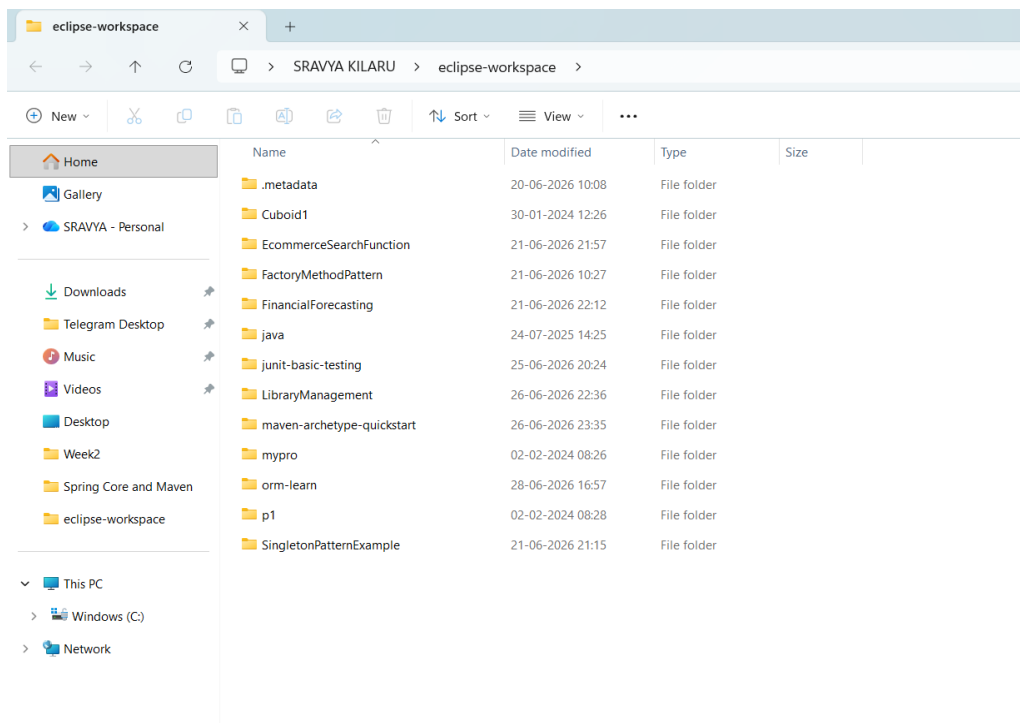
1. Spring Initializr Project Configuration

The Spring Boot project was configured using **Spring Initializr** with the required project settings and dependencies before generating the project.



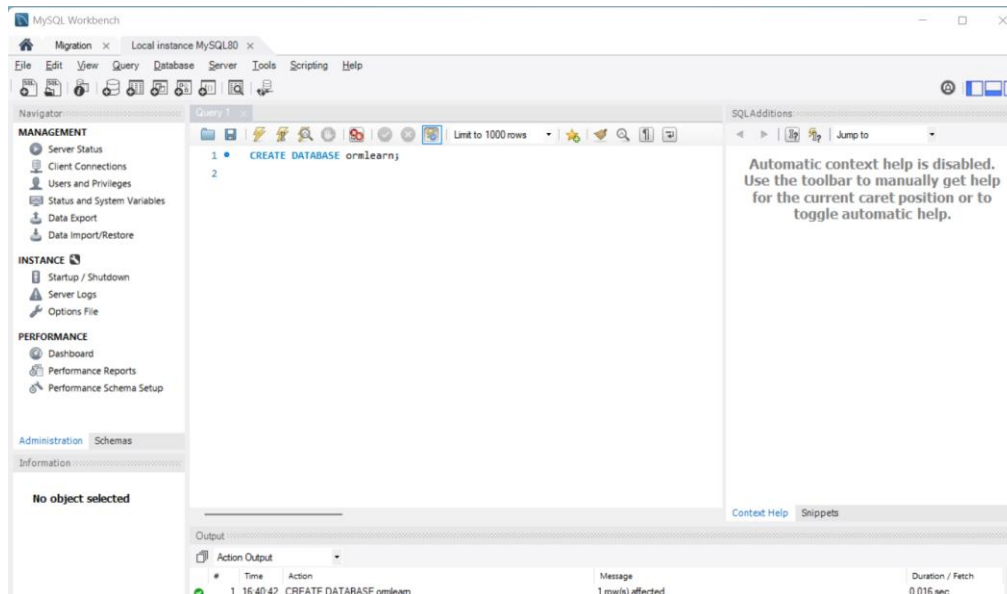
2. Generated Spring Boot Project Folder

After generating the Spring Boot project using Spring Initializr, the project folder **orm-learn** was successfully downloaded, extracted, and placed inside the Eclipse workspace (eclipse-workspace) for further development.



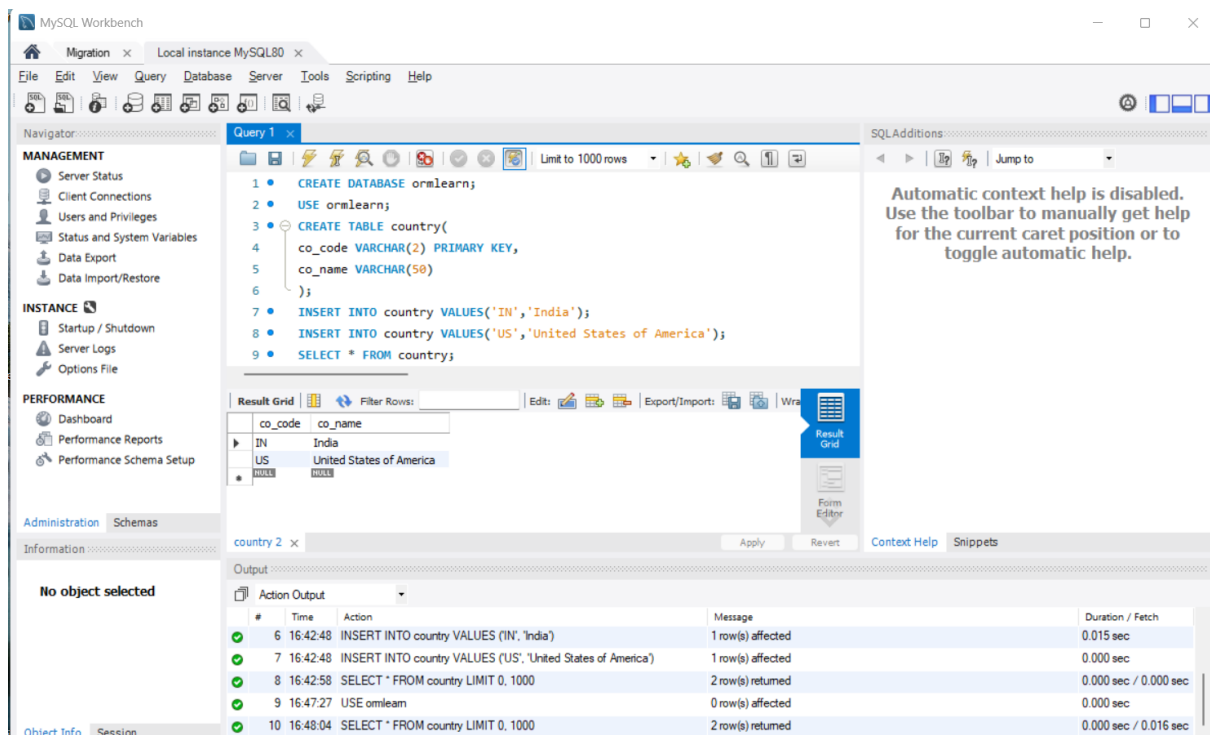
3. Creating MySQL Database

Created a new MySQL database named **ormlearn** using the **CREATE DATABASE SQL** command. This database will be used to store the application data for the Spring Data JPA project.



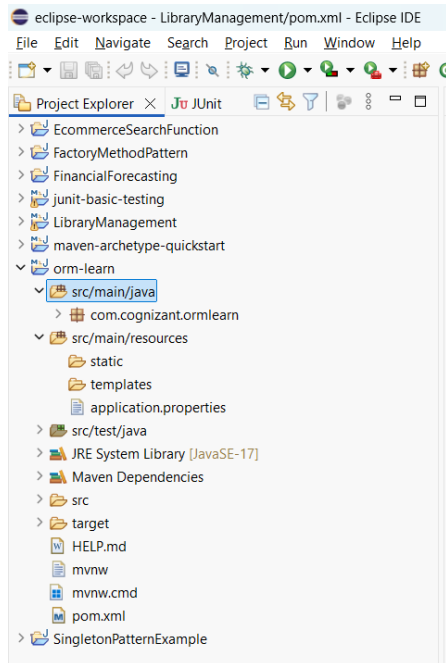
4. Creating Country Table and Inserting Sample Data

After creating the **ormlearn** database, the **country** table was created with **co_code** as the primary key and **co_name** as the country name. Two sample records were inserted into the table, and the data was verified using the **SELECT** query.



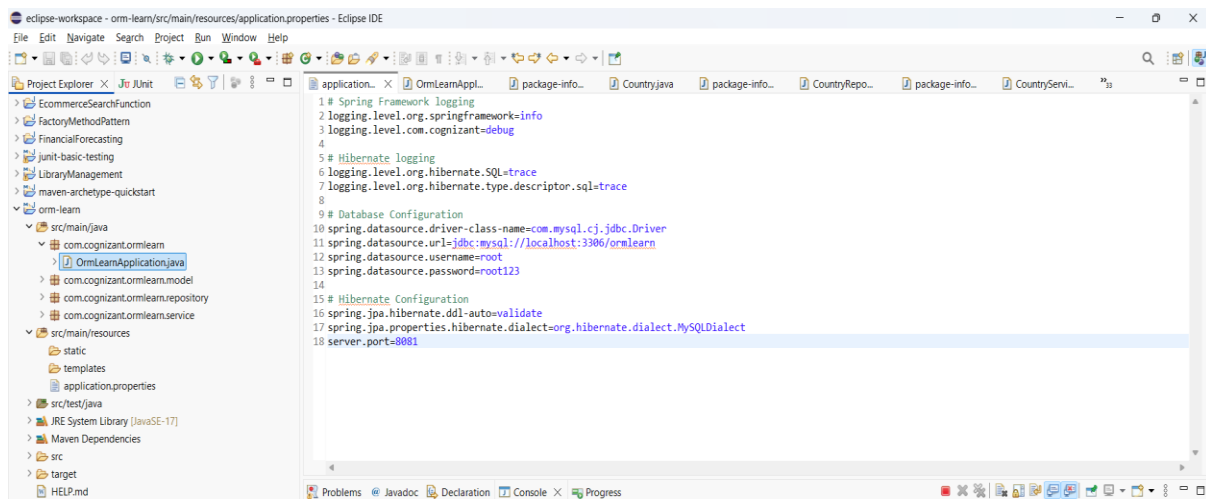
5. Importing the Maven Project into Eclipse

The generated **Spring Boot Maven project (orm-learn)** was successfully imported into the Eclipse IDE. The Project Explorer displays the project structure, including the Java source folders, resources, Maven dependencies, and the pom.xml file.



6. Database Configuration (application.properties)

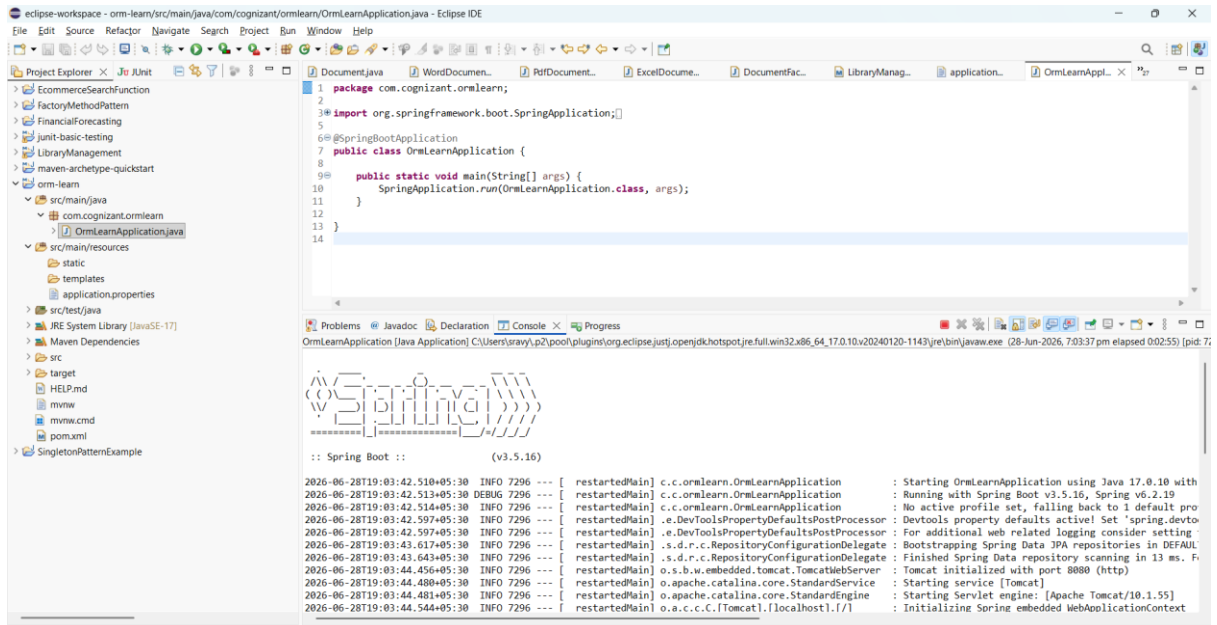
The **application.properties** file was configured to establish a connection between the Spring Boot application and the MySQL database. The database connection details, Hibernate properties, and logging levels were specified.



COGNIZANT DIGITAL NURTURE 5.0 – JAVA FSE

7. Executing the Spring Boot Application

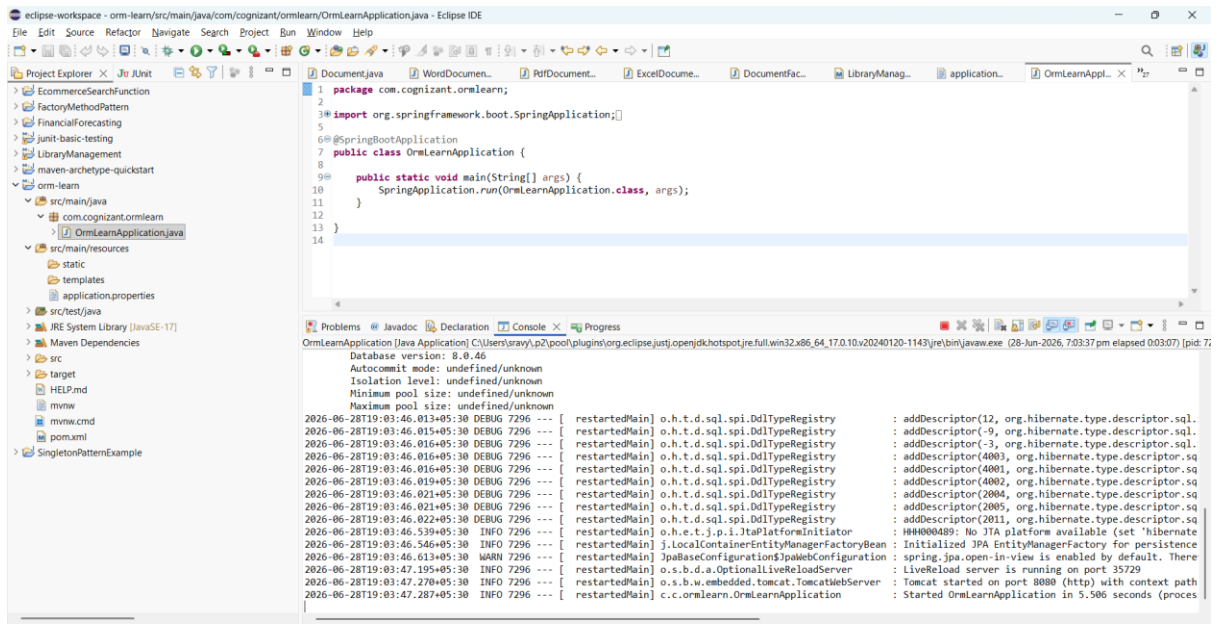
The **OrmLearnApplication.java** file is the main class of the Spring Boot project. The application was executed using the **Run As → Spring Boot App/Java Application** option. The console displays the Spring Boot startup logs, confirming that the application initialized successfully and established the connection with the MySQL database.



The screenshot shows the Eclipse IDE with the **OrmLearnApplication.java** file open. The code is as follows:

```
1 package com.cognizant.ormlearn;
2
3 import org.springframework.boot.SpringApplication;
4
5
6 @SpringBootApplication
7 public class OrmLearnApplication {
8
9     public static void main(String[] args) {
10         SpringApplication.run(OrmLearnApplication.class, args);
11     }
12 }
13
14
```

The console output shows the Spring Boot startup logs, including the version (v3.5.16) and the application running successfully. The logs also show the application's configuration, including the database version (8.0.46) and the application's startup time (28-Jun-2026, 7:03:37 pm).



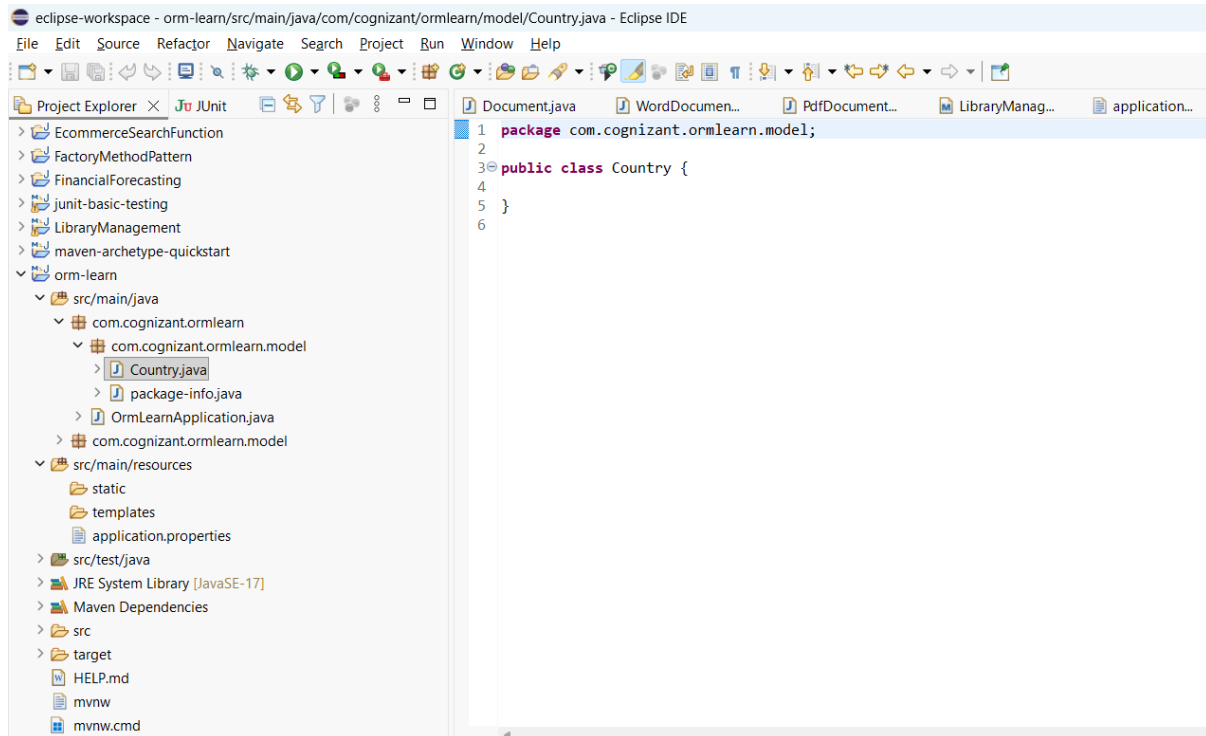
The screenshot shows the Eclipse IDE with the **OrmLearnApplication.java** file open. The code is as follows:

```
1 package com.cognizant.ormlearn;
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3 import org.springframework.boot.SpringApplication;
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The console output shows the Spring Boot startup logs, including the version (v3.5.16) and the application running successfully. The logs also show the application's configuration, including the database version (8.0.46) and the application's startup time (28-Jun-2026, 7:03:37 pm).

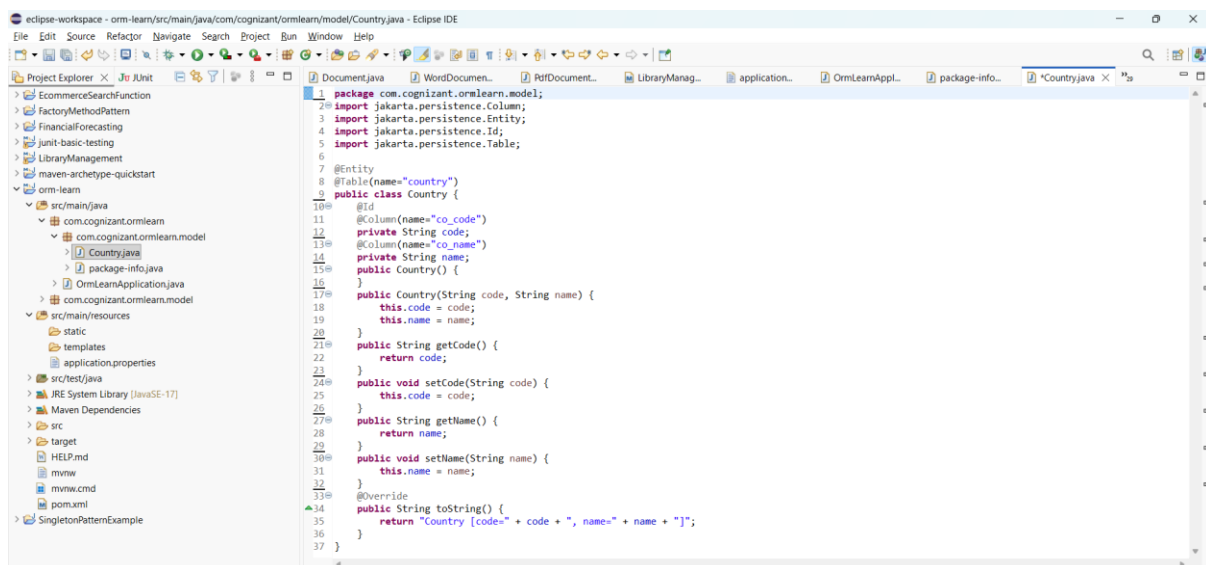
8. Creating the Country Entity Class

A new Java class named **Country.java** was created inside the **com.cognizant.ormlearn.model** package. This class represents the Country entity that will be mapped to the **country** table in the MySQL database using JPA annotations.



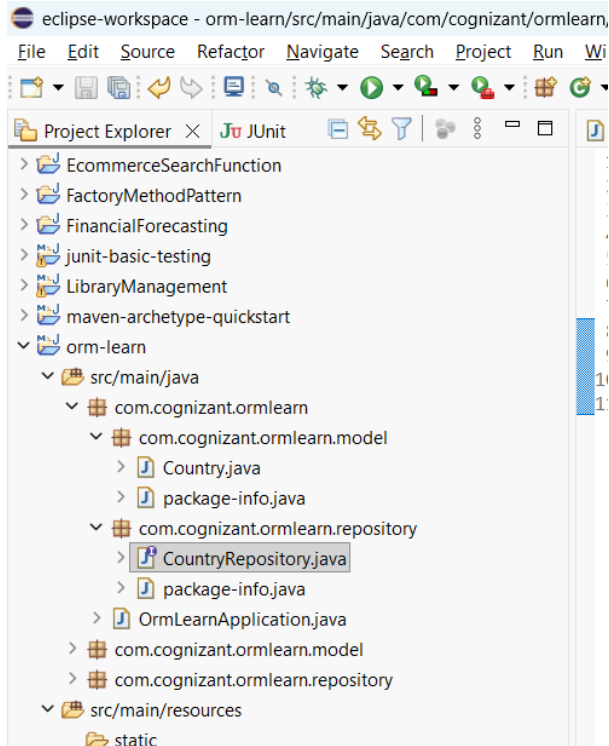
9. Implementing the Country Entity using JPA Annotations

The **Country.java** class was completed by adding JPA annotations and defining the entity attributes. The class was mapped to the **country** table in the MySQL database. Primary key, column mappings, constructors, getter and setter methods, and the **toString()** method were implemented.



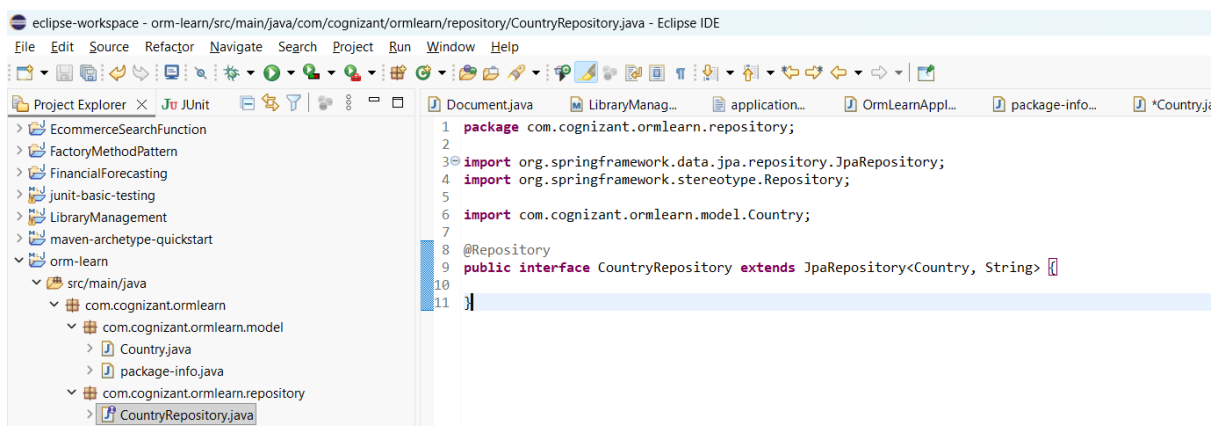
10. Creating the Country Repository Interface

A new repository package named **com.cognizant.ormlearn.repository** was created, and the **CountryRepository.java** interface was added. This repository provides database access operations for the Country entity using Spring Data JPA.



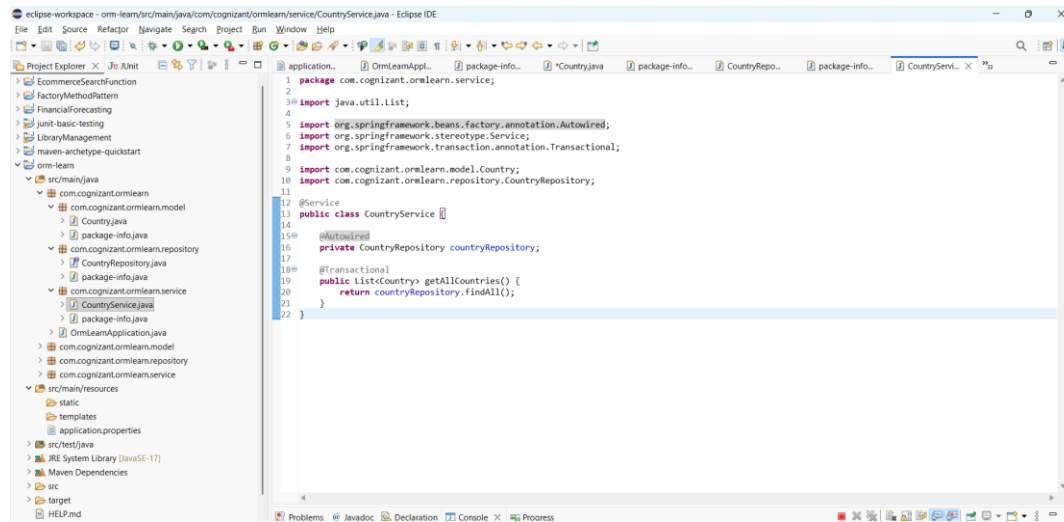
11. Implementing the Country Repository

The **CountryRepository** interface was implemented by extending **JpaRepository**. Spring Data JPA automatically provides built-in CRUD operations such as retrieving, inserting, updating, and deleting records without writing SQL queries.



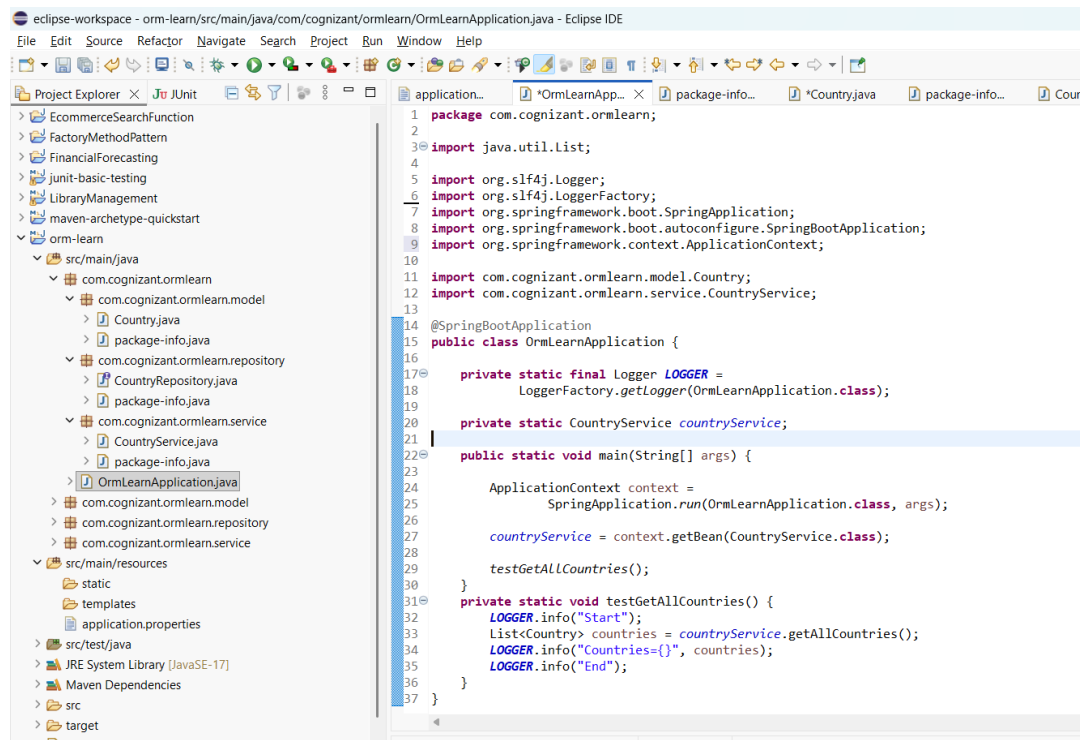
12. Creating the Service Layer (CountryService.java)

The **CountryService** class was created inside the **com.cognizant.ormlearn.service** package. This service layer contains the business logic and communicates with the repository layer to retrieve country records from the MySQL database.



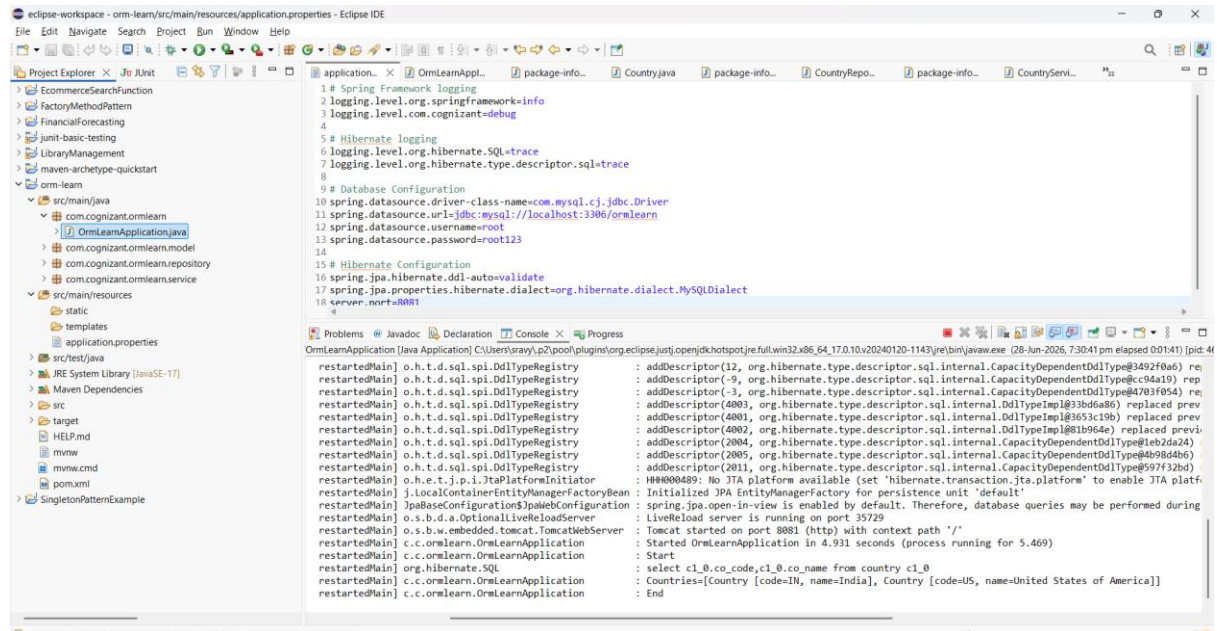
13. Updating the Main Application (OrmLearnApplication.java)

The **OrmLearnApplication.java** file was modified to invoke the service layer after the Spring Boot application starts. The application retrieves all country records from the MySQL database and displays them in the Eclipse Console using the Logger.



14. Successful Execution of the Spring Boot ORM Application

The Spring Boot ORM application was executed successfully after completing the project configuration. The console output confirms that the application connected to the MySQL database, initialized Hibernate ORM, started the embedded Tomcat server, executed the SQL query, and successfully retrieved the records from the **country** table.



```

1 # Spring Framework logging
2 logging.level.org.springframework.info
3 logging.level.com.cognizant-debug
4
5 # Hibernate logging
6 logging.level.org.hibernate.SQL=trace
7 logging.level.org.hibernate.type.descriptor.sql=trace
8
9 # Database Configuration
10 spring.datasource.driver-class-name=com.mysql.cj.jdbc.Driver
11 spring.datasource.url=jdbc:mysql://localhost:3306/ormlearn
12 spring.datasource.username=root
13 spring.datasource.password=root123
14
15 # Hibernate Configuration
16 spring.jpa.hibernate.ddl-auto=validate
17 spring.jpa.properties.hibernate.dialect=org.hibernate.dialect.MySQLDialect
18 server.port=8081
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